Document parsing project: what we tested, what we found, and what it means

A plain-language report. It starts from the first question and covers every kind of document we tested: digital PDFs, PDFs with tables, charts and photos, scans, Word, PowerPoint, Excel, CSV, plain text and images, then tough, real-world style and very long documents.

1. The whole story in one page

- **The question.** For a given document, which parser (the software that reads the file and turns it into text) gives the best result? And can the app pick that parser by itself, based on measured results, not on guesses?
- **What was wrong at the start.** The app chose parsers with fixed rules that nobody had measured. Some choices looked random, for example OpenDataLoader for almost every table document.
- **What we did.** (1) A fresh test: 62 documents, every parser on every document, each result scored against an answer file written by hand. (2) The app now picks parsers from those scores. (3) Then we tried to break it with 34 harder documents: tough tables, real-world style business documents, and long documents of 30 to 100 pages.
- **The main finding.** There is no best parser for everything. Plain text and simple tables are fine with fast parsers. Charts need the vision model (smart_pdf). Scans need OCR or a layout model (Docling). Merged cells and tables inside tables need a layout-aware parser (PyMuPDF4LLM, OpenDataLoader or Docling).
- **What still goes wrong.** A few documents lose a lot in Smart Extract (about 3.0 instead of 4.5 to 5.0). The causes are known and listed in section 8. Nothing in the routing has been changed because of the new tests.

Quick answer: which parser for which document

Document type	Best choice from the tests	Watch out for
Digital PDF, mostly text	PyMuPDF (fast, scores 4.9)	Nothing special.
Digital PDF with tables	OpenDataLoader, PyMuPDF4LLM or Docling (all about 4.5)	Merged cells and nested tables: plain PyMuPDF, hybrid_pdf and smart_pdf are weaker.
PDF with charts or photos	smart_pdf when the chart values matter	On documents with only photos or diagrams, the free parsers can score higher because the vision model adds descriptions.
Scanned PDF or photo of a page	Docling	Parsers without OCR return nothing. Tesseract alone loses table structure.
Image files (png, jpg)	Docling (4.85)	-
Word	Unstructured or Docling; the native reader is slightly lower	Native reader dropped the page header on one file.
PowerPoint	Native reader; Docling when the deck has real charts	Native reader cannot read chart values.
Excel / CSV	Native reader (CSV 5.0)	Merged cells and charts need checking.
Plain text	Plain text reader (5.0)	Do not send it to other parsers.

2. How we scored (in simple words)

Every parser output was compared with an answer file written by hand. The score is from 0 to 5 and is the average of up to eight questions:

Question	In simple words
Text	Are the words right?
Table	Are the rows and columns right?
Chart	Were the chart values read?
Infographic	Were the labels in a diagram read?
Numbers	Are the numbers, dates and names exact?
Structure	Are headings, lists and order kept?
Complete	Is anything missing?
No extras	Did the parser add things that are not there?

A crash, a timeout or an empty result counts as 0. Parsers within 0.1 of the best are treated as equal, and then the fastest one wins. Questions that do not apply to a document (for example Chart on a text page) are left out of its average.

3. The parsers and what they are

Parser	What it is	Good at	Weak at
PyMuPDF	Very fast reader for PDFs that have a text layer	Plain text, simple tables	Merged cells, nested tables, scans (returns nothing)
PyMuPDF4LLM	PyMuPDF plus layout rules that write Markdown	Merged headers, long reports	Slower on big files
OpenDataLoader	Layout-aware PDF reader	Tables, forms	Scans (nothing), some side-by-side tables
Docling	Layout model with OCR	Scans, tables, slides, images	Slow (minutes on long files)
Unstructured	General document splitter	Word, Excel	PDF tables, plain text
MarkItDown	Converts Office files to Markdown	Excel, CSV	PDF tables
pdfplumber	PDF text and table finder	Simple tables	Structure
Tesseract OCR	Reads text from pictures	Scans	Tables (becomes plain lines)
hybrid_pdf	PyMuPDF + Tesseract OCR	Mixed text and scanned pages	Same table weaknesses as PyMuPDF
smart_pdf	PyMuPDF + Tesseract OCR + a vision model that looks at charts, diagrams and photos (at most 8 per document)	Reading chart values	Merged-header tables; adds descriptions of photos
Native readers	Word, PowerPoint, Excel and text readers built into the app	Clean Office files	Charts inside Office files

Two parsers are kept out of the app on purpose: **vlm** (it sends every page to the Azure OpenAI model, too many API calls) and **Azure Document Intelligence** (paid; it can run next to the others as a comparison but was not part of these tests).

4. How Smart Extract works

Smart Extract looks at the document first, decides what kind it is, and picks parsers from the measured scores.

- **By document:** the whole file goes to the best parser for its type. If that parser fails, the next one is tried.
- **By part:** the file is split into pieces. Text and headings stay with PyMuPDF. Tables go to the parser that scored best on tables. Scanned pages go to the best scanned-page parser. Charts and photos go to the vision model (at most 8 per document).
- **Free or paid:** with *Include paid parsers* ticked, smart_pdf and the vision model can be used, and Azure Document Intelligence runs as an extra comparison tab. The result shown is always the best-ranked parser, never Azure by default.
- **Prefer quality:** turns off the tie rule, so the highest score wins even if it is slower.

5. First test: 62 documents

Every parser was run on every document it can open (786 runs). The table shows what Smart Extract uses first for each kind of document.

Document type	Documents tested	Free mode: first parser	Score	Paid allowed: first parser	Score
Digital text PDF	6	pymupdf	4.94	pymupdf	4.94
Digital PDF with tables	15	opendataloader	4.45	opendataloader	4.45
PDF with charts, figures or images	13	pymupdf4llm	3.29	smart_pdf	3.86
Mixed PDF (text pages and scanned pages)	3	pymupdf4llm	4.42	pymupdf4llm	4.42
Scanned PDF	8	docling	3.47	docling	3.47
Image files	4	docling	4.85	docling	4.85
Word	3	docx	4.10	docx	4.10
PowerPoint	2	pptx	4.86	pptx	4.86
PowerPoint with charts (real chart objects)	1	docling	4.79	docling	4.79
Excel	2	spreadsheet	4.33	spreadsheet	4.33
Excel with charts (real chart objects)	1	spreadsheet	5.00	spreadsheet	5.00
CSV	2	spreadsheet	5.00	spreadsheet	5.00
Plain text	2	text	5.00	text	5.00

Full parser-against-parser tables for all 13 document types are in Appendix A.

What the first test showed

- Digital text PDFs: PyMuPDF, hybrid_pdf and smart_pdf tie at 4.94, and PyMuPDF is the fastest.
- PDFs with tables: Docling 4.50, OpenDataLoader 4.45 and PyMuPDF4LLM 4.44 are tied within 0.1; plain PyMuPDF is lower (4.31).
- PDFs with charts or photos: smart_pdf is first (3.88) because it reads chart values; the next best is 3.32.
- Scans: Docling 3.47 is first; PyMuPDF and pdfminer return nothing.

- Images: Docling 4.85. Word 4.1 to 4.2 (all close). PowerPoint: native 4.86. Excel and CSV: native is at or near the top. Plain text: native 5.0.

Documents that mix text, tables, charts and photos

These were run through the real app, three ways.

These were run through the real app (By document with paid parsers allowed, By part with paid parsers allowed, and By document with free parsers only), with the final routing table. Azure Document Intelligence was blocked in this test. Score is 0 to 5 against the hand-written answer.

Document	By document (paid allowed)	By part (paid allowed)	By document (free only)
01_corporate_annual_report.pdf	2.53 (smart_pdf)	2.54 (pymupdf, vision:azure_openai, smart_pdf)	2.45 (pymupdf4llm)
02_invoice_purchase_order.pdf	3.75 (opendataloader)	3.75 (pymupdf, smart_pdf)	-
03_legal_contract.pdf	3.63 (opendataloader)	3.47 (pymupdf, smart_pdf)	-
04_scientific_research_paper.pdf	2.84 (smart_pdf)	2.84 (pymupdf, smart_pdf, vision:azure_openai)	3.19 (pymupdf4llm)
05_healthcare_discharge_summary.pdf	3.22 (opendataloader)	3.22 (pymupdf, smart_pdf)	-
06_insurance_claim_form.pdf	3.19 (opendataloader)	3.19 (pymupdf, smart_pdf)	-
07_logistics_shipping_manifest.pdf	3.57 (opendataloader)	3.33 (pymupdf, smart_pdf)	-
08_software_incident_report.pdf	3.18 (opendataloader)	3.18 (pymupdf, smart_pdf)	-
09_scanned_maintenance_order.pdf	5.00 (docling)	5.00 (docling)	-
10_mixed_parser_killer.pdf	2.23 (smart_pdf)	1.99 (pymupdf, vision:azure_openai, smart_pdf)	2.72 (pymupdf4llm)
annual_report_page.pdf	3.84 (smart_pdf)	3.84 (pymupdf, vision:azure_openai, smart_pdf)	3.25 (pymupdf4llm)
dashboard.pptx	4.80 (docling)	-	-
datasheet_with_photo.pdf	3.87 (smart_pdf)	3.87 (pymupdf, vision:azure_openai, smart_pdf)	4.94 (pymupdf4llm)
flowchart_doc.pdf	4.12 (smart_pdf)	4.12 (pymupdf, vision:azure_openai, smart_pdf)	4.40 (pymupdf4llm)
infographic.pdf	4.78 (docling)	4.78 (docling)	4.78 (docling)
product_brochure.pdf	4.11 (smart_pdf)	4.11 (pymupdf, vision:azure_openai, smart_pdf)	4.92 (pymupdf4llm)
report_with_figures.docx	3.44 (unstructured)	-	-
sales_with_chart.xlsx	5.00 (spreadsheet)	-	-
scanned_report_page.pdf	3.16 (docling)	3.16 (docling)	3.16 (docling)

Average score over these documents: By document: 3.70 (19 documents), By part: 3.52 (16 documents), By document, free only: 3.76 (9 documents).

What this shows. On the 9 documents that were run both ways, the free parsers scored higher on 5 (04_scientific_research_paper, 10_mixed_parser_killer, datasheet_with_photo, flowchart_doc, product_brochure) and the paid vision path on 2 (01_corporate_annual_report, annual_report_page). The vision model clearly helps when the values only exist inside a chart, and it costs points when the document is mostly text with photos or diagrams, because the model adds descriptions that are not in the answer. The "charts and figures" routing group mixes both kinds, so its first pick (smart_pdf when paid parsers are allowed) is right on average but not on every document.

Reading the scores. The ten Copilot documents (names starting 01 to 10) score lower for every parser, because their answer files are summaries and not the full text. Compare parsers on the same document, not documents with each other.

6. The three new test sets

All three sets were generated by scripts in this project (`make_enterprise_docs.py`, `make_realistic_docs.py`, `make_long_docs.py`), each with a hand-written answer file. **None of them are real company files**; real files cannot be shared here. The layouts copy common real formats.

Set	Documents	What it contains
Tough documents	14	merged-cell tables, a table inside a table, a table over many pages, wide tables, forms, a scan, charts with photos, Excel and Word
Real-world style	14	bank statement, income statement, purchase order, payslip, lab report, electricity bill with chart, bill of lading, RFP matrix, board minutes, two-column paper, a tilted phone photo of a purchase order, Excel with chart, Word SOP, PowerPoint with chart
Long documents	6	56-page annual report with 10 charts, 100-page ledger (2,500 rows), 46-page contract, 31 scanned pages, 40-section Word handbook, 5-sheet Excel workbook

How the three sets did, in numbers

Smart Extract average score (0 to 5) over the PDFs in each set:

Set	PDFs	By document, paid	By part, paid	By document, free
Tough documents	12	4.32 (12)	4.05 (12)	4.59 (12)
Real-world style documents	11	4.59 (11)	4.26 (11)	4.44 (11)
Long documents	0	-	-	-

The number in brackets is how many documents finished. A smaller number means some runs are still going or did not finish.

7. Chapter by document type

Digital PDFs (mostly text)

The easy case. Every parser with a text layer scores well, so speed decides. On the long contract and long report, PyMuPDF4LLM and Docling lead; the PyMuPDF family is a little lower.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
`contract_nested_clauses.pdf`	Tough documents	3	Docling 4.67	4.41 / 4.16 / 4.41	Docling keeps the clause numbering best.
`long_annual_report_20p.pdf`	Tough documents	19	PyMuPDF4LLM 5.00 (tied: Docling)	4.89 / 4.49 / 4.89	PyMuPDF4LLM and Docling are perfect (5.0).
`board_minutes.pdf`	Real-world style documents	1	PyMuPDF4LLM 5.00 (tied: Docling, PyMuPDF, hybrid_pdf)	4.61 / 4.91 / 4.61	Mostly text. PyMuPDF4LLM and Docling are perfect. By document picked OpenDataLoader (4.61) because its rule is per document type, not per file.

contract_45p.pdf	Long documents	46	PyMuPDF4LLM 4.99 (tied: PyMuPDF, hybrid_pdf)	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

Digital PDFs with tables, forms and statements

The biggest group. Simple tables are easy for every layout-aware parser. Merged cells, side-by-side tables, nested tables and header rows that span columns are where parsers separate. The same few parsers (PyMuPDF4LLM, OpenDataLoader, Docling) keep leading, and plain PyMuPDF, hybrid_pdf and smart_pdf trail on the hard ones.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
bom_wide_table.pdf	Tough documents	1	Docling 5.00 (tied: OpenDataLoader, PyMuPDF, hybrid_pdf)	4.99 / 4.89 / 4.99	12 columns in small print. The top parsers all keep every column.
clinical_trial_table.pdf	Tough documents	1	PyMuPDF4LLM 5.00	3.01 / 3.02 / 5.00	PyMuPDF4LLM scores 5.0. smart_pdf scores 3.01 because the picture detector mistook the table lines for a figure.
financial_statement_multiheader.pdf	Tough documents	1	PyMuPDF4LLM 4.86	2.96 / 3.12 / 4.86	Two-row merged header. PyMuPDF4LLM and Docling keep it together. Smart Extract with paid parsers fell to 2.96 because it copied the same numbers twice (see section 7).
form_grid_ac4.pdf	Tough documents	1	PyMuPDF 4.55 (tied: hybrid_pdf, smart_pdf, OpenDataLoader)	4.52 / 4.42 / 4.52	Easy for the PyMuPDF family. Nothing special.
invoice_complex.pdf	Tough documents	1	OpenDataLoader 4.85	4.47 / 4.38 / 4.67	OpenDataLoader is best. smart_pdf adds a description of the QR code that is not in the answer.
ledger_repeated_header.pdf	Tough documents	3	Docling 5.00 (tied: OpenDataLoader)	4.95 / 4.86 / 4.95	The header repeating on every page did not hurt anybody.
policy_schedule_nested.pdf	Tough documents	1	OpenDataLoader 4.77 (tied: PyMuPDF4LLM)	4.77 / 3.06 / 4.77	A table inside a table. PyMuPDF4LLM and OpenDataLoader keep it right. PyMuPDF-based parsers repeat the outer text across columns.
annual_report_income_statement.pdf	Real-world style documents	1	OpenDataLoader 4.77 (tied: PyMuPDF4LLM)	4.77 / 4.25 / 4.77	Dollar signs in their own cells and numbers in brackets. Layout-aware parsers (OpenDataLoader, PyMuPDF4LLM, Docling) beat plain PyMuPDF by 0.3 to 0.4.
bank_statement.pdf	Real-world style documents	3	PyMuPDF4LLM 4.77 (tied: OpenDataLoader, Docling, PyMuPDF)	4.76 / 4.68 / 4.76	3 pages, header repeated. Every parser with a text layer does well (4.5 to 4.8).
bill_of_lading.pdf	Real-world style documents	1	OpenDataLoader 4.56	4.56 / 3.93 / 4.56	Merged cells. OpenDataLoader, Docling and PyMuPDF4LLM are ahead of the PyMuPDF family (3.85).
lab_report.pdf	Real-world style documents	1	Docling 4.93 (tied: PyMuPDF4LLM, PyMuPDF, hybrid_pdf)	4.55 / 3.26 / 4.55	Docling, PyMuPDF4LLM and the PyMuPDF family all above 4.8. By part scored only 3.26: the table piece came out with every merged cell repeated across 12 columns.

payslip.pdf	Real-world style documents	1	Docling 4.69 (tied: PyMuPDF4LLM)	3.94 / 3.86 / 3.94	Two tables side by side. Docling and PyMuPDF4LLM are best (4.6 to 4.7); OpenDataLoader, the choice for this type, scored 3.94.
purchase_order.pdf	Real-world style documents	1	OpenDataLoader 4.96	4.96 / 3.54 / 4.96	OpenDataLoader 4.96 is best. By part scored only 3.54.
rfp_requirements_matrix.pdf	Real-world style documents	1	Docling 5.00 (tied: PyMuPDF, hybrid_pdf, smart_pdf)	4.75 / 4.92 / 4.75	Rowspan category cells and check boxes were easy: every top parser is 4.75 or better.
transaction_ledger_100p.pdf	Long documents	100	OpenDataLoader 5.00 (tied: PyMuPDF4LLM)	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

PDFs with charts, figures and photos

Only a vision model can read the numbers inside a chart. smart_pdf is first on each of the four chart documents of the tough and real-world sets (annual report spread, mixed page, research paper, electricity bill). The cost is a few seconds per figure and one API call per figure.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
annual_report_spread.pdf	Tough documents	2	smart_pdf 4.53	4.53 / 3.96 / 4.05	Chart page. smart_pdf wins because it reads the chart values.
mixed_table_chart_photo.pdf	Tough documents	1	smart_pdf 3.90	3.90 / 3.79 / 3.55	Table + chart + photo on one page. smart_pdf is best because it reads the chart.
research_two_column.pdf	Real-world style documents	1	smart_pdf 4.43	4.43 / 4.32 / 3.74	Two columns with a chart. smart_pdf is clearly best (4.43) because it reads the chart.
utility_bill.pdf	Real-world style documents	1	smart_pdf 4.49	4.49 / 4.50 / 3.58	Bill with a bar chart. smart_pdf wins (4.49) because it reads the chart; the free route scores 3.58.
annual_report_60p.pdf	Long documents	56	PyMuPDF4LLM 4.18	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

Scanned PDFs and photos of pages

A scan has no text layer. Parsers that only read text return nothing (counted as 0). Docling reads scans best. hybrid_pdf and smart_pdf use only Tesseract OCR on scanned pages, so tables become plain lines.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
scanned_financial_statement.pdf	Tough documents	1	PyMuPDF4LLM 4.49 (tied: Docling)	4.41 / 4.41 / 4.41	A scan: parsers without OCR return nothing. Docling and PyMuPDF4LLM read it.
scanned_purchase_order_phone.pdf	Real-world style documents	1	Docling 4.67	4.67 / 4.67 / 4.67	A tilted, noisy phone photo. Docling 4.67 is far ahead; hybrid_pdf and smart_pdf (Tesseract only) get 2.30.
scanned_report_30p.pdf	Long documents	31	Docling 4.58	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

Word documents

Native Word reader is close to the others but lost the page header and flattened the list numbers on both Word files; Unstructured scored highest on both.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
`quality_manual.docx`	Tough documents	Word	Unstructured 4.86	4.09 / - / -	Unstructured is clearly better than the native Word reader (the native one drops the page header).
`sop_goods_receipt.docx`	Real-world style documents	Word	Unstructured 4.93	4.33 / - / -	Unstructured (4.93) is ahead of Docling (4.57) and the native Word reader (4.33).
`handbook_40p.docx`	Long documents	Word	native Word reader 5.00 (tied: Unstructured, Docling, MarkItDown)	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

PowerPoint

Native reader is best for normal slides. For a deck that has real chart objects, Docling reads the chart and the native reader cannot.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
`quarterly_review.pptx`	Real-world style documents	slides	Docling 4.50	4.50 / - / -	Deck with a real chart. Docling reads the chart (4.50); the native reader does not (4.01).

Excel

Native reader is close to the top. MarkItDown and Unstructured are equal or slightly better on some files, but not by the 0.5 margin the rule needs to replace the native reader.

Document	Set	Pages	Best parser	Smart Extract: By document, By part, free	What happened
`budget_model.xlsx`	Tough documents	Excel	MarkItDown 4.31	4.04 / - / -	MarkItDown is slightly better than the native Excel reader; the rule keeps native first unless another parser wins by more than 0.5.
`hr_headcount_report.xlsx`	Real-world style documents	Excel	Unstructured 3.75 (tied: native Excel reader)	3.73 / - / -	Excel with a real chart object. Native reader and Unstructured are tied (3.7).
`big_workbook.xlsx`	Long documents	Excel	native Excel reader 4.99	- / - / -	Result is in the ranking table in Appendix B; see the note below the table.

8. The problems we found

These are the real weak points. None of them is fixed yet, because you asked for testing, not changes.

1. **Table lines mistaken for a figure (clinical_trial_table 3.01, financial_statement_multiheader 2.96 in paid By document; free route 5.0 and 4.86).** PyMuPDF splits a table with merged cells into several small tables. The picture detector then sees one big group of lines that is not inside any single table and calls it a figure. We confirmed this by running the detector directly. The file is then sent to smart_pdf, which reads the table as a picture and also adds the PyMuPDF text, so the numbers appear twice. Fix idea: also check figures against the combined area of all detected tables.

2. **By part repeats merged cells (purchase_order 3.54 vs 4.96 By document; lab_report 3.26 vs 4.55; bill_of_lading 3.93 vs 4.56; policy_schedule_nested 3.06 vs 4.77).** In By part, a table piece can be read by PyMuPDF, which writes the same text into every column a merged cell covers. We saw this directly in the lab_report output; the same pattern is probable on the others. Fix idea: keep tables with merged cells or nested tables as one piece and give them to PyMuPDF4LLM or OpenDataLoader.
3. **A network error looks like a bad score.** annual_report_spread first scored 3.98 in Smart Extract. The vision call for the bar chart had failed with a connection error, so the chart was missing. Run again, it scored 4.53, the same as smart_pdf alone. Lesson: when a figure cannot be read, the result shows a warning line, and such a run should be repeated.
4. **The rule is per document type, not per file.** For digital PDFs with tables the first choice is OpenDataLoader. On board_minutes (mostly text) it scored 4.61 while PyMuPDF4LLM and Docling scored 5.0, and on payslip it scored 3.94 while Docling scored 4.69. A per-file check would do better but costs a test run per file.
5. **hybrid_pdf and smart_pdf are weak on scans.** Both use only Tesseract OCR on scanned pages: 2.3 to 2.4 on the two scanned documents of the tough and real-world sets, against 4.4 to 4.7 for Docling. Smart Extract avoids this by sending scans to Docling.
6. **Native Word and Excel readers lose a little.** The native Word reader drops the page header and flattens numbered lists. The difference is under the 0.5 margin on most files.

9. Long documents

Long documents were run with a limit of 10 minutes per parser per document (parsers vs parsers) and 15 minutes per Smart Extract run. A parser that does not finish is counted as 0 and shown as *timeout*.

Document	Pages	Best parser	Parsers that timed out or returned nothing	Smart Extract: By document, By part, free
`annual_report_60p.pdf`	56	PyMuPDF4LLM 4.18	Docling (timeout)	- / - / -
`big_workbook.xlsx`	Excel	native Excel reader 4.99	none	- / - / -
`contract_45p.pdf`	46	PyMuPDF4LLM 4.99 (tied: PyMuPDF, hybrid_pdf)	none	- / - / -
`handbook_40p.docx`	Word	native Word reader 5.00 (tied: Unstructured, Docling, MarkItDown)	none	- / - / -
`scanned_report_30p.pdf`	31	Docling 4.58	pdfplumber (empty), PyMuPDF (empty), MarkItDown (empty), OpenDataLoader (empty)	- / - / -
`transaction_ledger_100p.pdf`	100	OpenDataLoader 5.00 (tied: PyMuPDF4LLM)	none	- / - / -

Smart Extract runs finished for long documents so far: 0. If this number is lower than 14, the run is still going and this section will be updated when the report is generated again.

Timings for long documents are measured but not exact, because the laptop was doing other things at the same time. What they show clearly: Docling and Tesseract OCR take minutes on 30 to 100 pages, while PyMuPDF takes seconds.

10. What is done and what is not

Done and tested

- Fresh 62-document head-to-head, with the routing built from it.
- Smart Extract By document, By part and free mode, in the app, on all three new sets.

- vlm is off everywhere; smart_pdf and By part read figures only (at most 8 per document).
- Detector problem and network error problem traced to their causes.

Not done

- Azure Document Intelligence on the new sets (it is paid; ask first).
- Fixes for the six problems in section 8. They change routing, so they need your yes.
- Real client documents: everything here is generated.
- The new sets are not part of the routing table. They only measure.

11. Limits

- All documents are generated; real files are messier.
- One run per parser and document. Differences under 0.1 are noise.
- Answer files are written by hand; a different but correct layout can lose points.
- Timings are only good for telling fast parsers from slow ones.
- LlamaParse (no key), MinerU, Marker and Azure Document Intelligence were not run.

Appendix A. Parser against parser, the 62-document test

Higher is better. A crash or a timeout counts as 0 on every parameter the document can test. Seconds are the median per document.

Digital text PDF (6 documents)

Documents: AI_agent_p1-3, contract, form_fields, multilingual, resume, rotated_pages.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
pymupdf	4.94	4.97	-	-	4.77	0.0	0	6
hybrid_pdf (PyMuPDF + Tesseract OCR)	4.94	4.97	-	-	4.77	0.1	0	6
smart_pdf (PyMuPDF + Tesseract OCR + a vision model for charts and figures) (paid)	4.94	4.97	-	-	4.77	0.1	0	6
opendataloader	4.75	4.80	-	-	4.48	2.7	0	6
haystack_converter	4.63	4.97	-	-	3.22	0.1	0	6
langchain_loader	4.63	4.97	-	-	3.22	0.3	0	6
llamaindex_reader	4.63	4.97	-	-	3.22	0.4	0	6
pymupdf4llm	4.60	4.58	-	-	4.42	1.5	0	6
pdfplumber	4.47	4.80	-	-	2.97	0.1	0	6
... 6 more parsers scored lower								

Digital PDF with tables (15 documents)

Documents: 02_invoice_purchase_order, 03_legal_contract, 05_healthcare_discharge_summary, 06_insurance_claim_form, 07_logistics_shipping_manifest, 08_software_incident_report, balance_sheet, bank_statement, datasheet, digital, invoice, lab_report, multipage_table, purchase_order, two_column_paper.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
docling	4.50	4.45	4.99	-	4.22	27.5	0	15
opendataloader	4.45	4.37	4.97	-	4.15	2.7	0	15
pymupdf4llm	4.44	4.37	4.96	-	4.17	2.8	0	15
pymupdf	4.31	4.23	4.77	-	3.98	0.2	0	15
hybrid_pdf (PyMuPDF + Tesseract OCR)	4.31	4.23	4.77	-	3.98	0.4	0	15
smart_pdf (PyMuPDF + Tesseract OCR + a vision model for charts and figures) (paid)	4.31	4.23	4.77	-	3.98	0.3	0	15
pdfplumber	4.10	4.35	4.84	-	2.30	0.2	0	15
markitdown	3.76	4.29	2.56	-	2.68	0.5	0	15
haystack_converter	3.37	4.36	0.00	-	2.73	0.1	0	15
... 6 more parsers scored lower								

PDF with charts, figures or images (13 documents)

Documents: 01_corporate_annual_report, 04_scientific_research_paper, 10_mixed_parser_killer, Generative_AI_p1-6, Overview of Amazon Web Services - Overview of Amazon Web Services, PIB_home_p1-6, annual_report_page, chart_report, chart_sample, datasheet_with_photo, flowchart_doc, product_brochure, slides.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
smart_pdf (PyMuPDF + Tesseract OCR + a vision model for charts and figures) (paid)	3.88	3.26	4.54	4.83	2.72	0.5	0	13
pymupdf4llm	3.32	3.92	3.94	1.07	3.33	7.4	0	13
opendataloader	3.17	3.83	3.94	0.60	3.24	3.4	0	13
pymupdf	3.02	3.63	3.80	0.60	3.02	0.3	0	13
hybrid_pdf (PyMuPDF + Tesseract OCR)	3.02	3.63	3.80	0.60	3.02	0.3	0	13
pdfplumber	3.01	3.84	3.96	0.60	1.98	0.3	0	13
docling	2.92	3.65	3.94	0.60	3.41	28.7	0	13
markitdown	2.53	3.67	1.10	0.00	2.04	0.7	0	13
haystack_converter	2.52	3.78	0.00	0.60	2.12	0.2	0	13
... 6 more parsers scored lower								

Mixed PDF (text pages and scanned pages) (3 documents)

Documents: AI_agent_mixed, mixed, mixed_digital_scanned.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
docling	4.51	4.30	5.00	-	4.00	42.5	0	3

pymupdf4llm	4.42	4.30	4.55	-	4.00	8.0	0	3
hybrid_pdf (PyMuPDF + Tesseract OCR)	3.83	4.20	2.50	-	3.33	4.3	0	3
smart_pdf (PyMuPDF + Tesseract OCR + a vision model for charts and figures) (paid)	3.83	4.20	2.50	-	3.33	9.6	0	3
easyocr	3.32	4.30	0.00	-	2.07	64.1	0	3
ocr	3.11	4.13	0.00	-	2.07	8.4	0	3
pymupdf	2.99	3.03	2.50	-	2.77	0.1	0	3
opendataloader	2.98	3.03	2.50	-	2.70	2.7	0	3
pdfplumber	2.79	3.07	2.50	-	1.53	0.1	0	3
... 6 more parsers scored lower								

Scanned PDF (8 documents)

Documents: 09_scanned_maintenance_order, AI_agent_scanned_clean, AI_agent_scanned_noisy, infographic, scanned_invoice_clean, scanned_invoice_low_quality, scanned_invoice_skewed_noisy, scanned_report_page.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
docling	3.47	4.21	3.10	0.00	3.80	37.6	0	8
pymupdf4llm	3.25	4.24	2.05	0.00	3.76	10.2	0	8
unstructured	2.88	4.04	0.00	1.20	3.44	15.5	0	8
easyocr	2.70	4.19	0.00	0.00	2.48	68.4	0	8
hybrid_pdf (PyMuPDF + Tesseract OCR)	2.53	3.90	0.00	0.00	2.56	3.0	0	8
ocr	2.53	3.90	0.00	0.00	2.56	3.2	0	8
smart_pdf (PyMuPDF + Tesseract OCR + a vision model for charts and figures) (paid)	2.53	3.90	0.00	0.00	2.56	7.5	0	8
pymupdf	0.00	0.00	0.00	0.00	0.00	0.0	0	8
pdfminer	0.00	0.00	0.00	0.00	0.00	0.0	0	8
... 6 more parsers scored lower								

Image files (4 documents)

Documents: handwritten_note.png, id_card.png, receipt_photo.jpg, screenshot.png.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
docling	4.85	4.85	-	-	4.90	12.6	0	4
easyocr	4.68	4.75	-	-	4.55	8.2	0	4
ocr	4.51	4.50	-	-	4.75	0.6	0	4
unstructured	4.44	4.47	-	-	4.60	3.7	0	4

Word (3 documents)

Documents: procedure.docx, report.docx, report_with_figures.docx.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
unstructured	4.20	4.60	4.10	2.50	4.53	0.4	0	3
docling	4.17	4.60	3.90	2.50	4.53	0.8	0	3
docx	4.10	4.53	3.90	2.50	4.53	0.1	0	3
markitdown	3.96	4.47	4.10	2.50	3.53	0.6	0	3

PowerPoint (2 documents)

Documents: deck.pptx, slides.pptx.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
pptx	4.86	4.75	5.00	-	4.85	0.0	0	2
docling	4.62	4.30	5.00	-	4.65	0.3	0	2
unstructured	4.56	4.30	5.00	-	4.25	0.2	0	2
markitdown	4.51	4.75	5.00	-	4.35	0.2	0	2

PowerPoint with charts (real chart objects) (1 documents)

Documents: dashboard.pptx.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
docling	4.79	4.80	4.80	5.00	4.30	1.3	0	1
markitdown	3.84	4.00	3.40	5.00	3.70	0.7	0	1
pptx	2.76	3.30	2.70	0.00	3.70	0.0	0	1
unstructured	2.76	3.30	2.70	0.00	3.70	0.4	0	1

Excel (2 documents)

Documents: multi_sheet.xlsx, sales.xlsx.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
unstructured	4.45	4.50	4.65	-	4.15	1.4	0	2
spreadsheet	4.33	4.40	4.40	-	4.20	0.0	0	2
markitdown	4.33	4.35	4.50	-	4.15	0.2	0	2
docling	4.26	4.45	4.45	-	3.50	0.4	0	2

Excel with charts (real chart objects) (1 documents)

Documents: sales_with_chart.xlsx.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
spreadsheet	5.00	5.00	5.00	-	5.00	0.1	0	1
markitdown	5.00	5.00	5.00	-	5.00	0.5	0	1
unstructured	4.43	4.80	5.00	-	2.30	2.4	0	1
docling	3.05	3.10	3.40	-	1.60	1.0	0	1

CSV (2 documents)

Documents: parts.csv, sales.csv.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
spreadsheet	5.00	5.00	5.00	-	5.00	0.8	0	2
markitdown	5.00	5.00	5.00	-	5.00	1.8	0	2
unstructured	5.00	5.00	5.00	-	5.00	2.2	0	2

Plain text (2 documents)

Documents: runbook.md, shift_notes.txt.

Parser	Overall	Text	Tables	Charts	Structure	Seconds	Failed	Runs
text	5.00	5.00	5.00	-	5.00	0.0	0	2
unstructured	2.08	2.50	0.00	-	2.50	0.1	1	2

Appendix B. Ranking per document: tough, real-world and long sets

Sorted by score. *tied* means within 0.1 of the best. – means the question does not apply.

annual_report_spread.pdf (Tough documents, 2 pages)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	smart_pdf	**4.53**	4.60	4.10	5.00	4.80	4.10	4.90	4.30	5
2	PyMuPDF4LLM	**4.05**	4.70	3.60	2.10	4.30	4.30	4.60	4.70	74
3	PyMuPDF	**3.69**	4.60	3.80	0.00	4.10	4.20	4.20	4.90	2
4	hybrid_pdf	**3.69**	4.60	3.80	0.00	4.10	4.20	4.20	4.90	0
5	OpenDataLoader	**3.50**	4.30	3.90	0.00	3.60	4.10	3.70	5.00	19
6	Docling	**3.42**	4.10	4.00	0.00	3.40	3.90	3.50	5.00	189
7	pdfplumber	**3.40**	4.60	3.90	0.00	4.10	2.00	4.20	5.00	3
8	MarkItDown	**3.07**	4.60	1.40	0.00	4.10	2.10	4.20	5.00	2

9	Tesseract OCR	**2.84**	4.20	0.00	2.10	3.10	2.10	3.80	4.50	4
10	Unstructured	**2.81**	4.20	0.00	0.00	4.10	2.70	3.80	4.80	17

Note: Chart page. smart_pdf wins because it reads the chart values.

bom_wide_table.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	82
2	OpenDataLoader (tied)	**4.99**	5.00	5.00	5.00	5.00	5.00	5.00	20
3	PyMuPDF (tied)	**4.93**	4.90	4.90	5.00	5.00	5.00	4.90	14
4	hybrid_pdf (tied)	**4.93**	4.90	4.90	5.00	5.00	5.00	4.90	3
5	smart_pdf (tied)	**4.93**	4.90	4.90	5.00	5.00	5.00	4.90	2
6	PyMuPDF4LLM	**4.77**	5.00	3.60	5.00	5.00	5.00	5.00	9
7	pdfplumber	**4.53**	4.90	4.90	5.00	2.50	5.00	4.90	14
8	MarkItDown	**3.54**	5.00	0.00	5.00	1.30	5.00	5.00	4
9	Unstructured	**3.54**	5.00	0.00	5.00	1.30	5.00	5.00	1
10	Tesseract OCR	**2.90**	3.90	0.00	3.60	1.90	3.30	4.80	4

Note: 12 columns in small print. The top parsers all keep every column.

budget_model.xlsx (Tough documents)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	MarkItDown	**4.31**	4.60	4.10	4.00	4.10	4.60	4.60	3
2	native Excel reader	**4.04**	4.30	3.80	3.50	3.90	4.60	4.00	8
3	Unstructured	**3.97**	4.30	3.80	3.50	3.50	4.60	4.00	4
4	Docling	**3.71**	4.20	3.80	3.50	2.20	4.60	3.90	187

Note: MarkItDown is slightly better than the native Excel reader; the rule keeps native first unless another parser wins by more than 0.5.

clinical_trial_table.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	6
2	Docling	**4.82**	4.70	5.00	5.00	4.80	4.50	5.00	367

3	OpenDataLoader	**4.43**	4.90	4.50	5.00	2.50	4.90	4.80	18
4	pdfplumber	**4.26**	4.90	4.10	4.90	2.00	5.00	4.70	1
5	PyMuPDF	**4.08**	4.60	3.60	4.80	2.20	4.90	4.30	2
6	hybrid_pdf	**4.08**	4.60	3.60	4.80	2.20	4.90	4.30	1
7	MarkItDown	**3.66**	4.90	0.00	5.00	2.20	4.90	4.90	1
8	Unstructured	**3.56**	4.90	0.00	5.00	1.60	4.90	4.90	0
9	Tesseract OCR	**3.35**	4.50	0.00	4.30	2.30	4.10	4.90	4
10	smart_pdf	**3.01**	3.20	2.70	3.20	1.60	4.90	2.40	0

Note: PyMuPDF4LLM scores 5.0. smart_pdf scores 3.01 because the picture detector mistook the table lines for a figure.

contract_nested_clauses.pdf (Tough documents, 3 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**4.67**	4.70	5.00	4.10	4.80	4.40	5.00	108
2	PyMuPDF4LLM	**4.45**	5.00	2.00	5.00	4.70	5.00	5.00	12
3	OpenDataLoader	**4.41**	4.50	5.00	4.10	3.60	4.30	4.90	20
4	PyMuPDF	**4.21**	4.90	1.70	5.00	4.10	4.90	4.80	5
5	hybrid_pdf	**4.21**	4.90	1.70	5.00	4.10	4.90	4.80	1
6	smart_pdf	**4.21**	4.90	1.70	5.00	4.10	4.90	4.80	0
7	pdfplumber	**3.96**	5.00	1.70	5.00	2.20	5.00	4.90	2
8	Unstructured	**3.70**	4.90	0.00	5.00	2.60	4.90	4.90	0
9	Tesseract OCR	**3.67**	5.00	0.00	5.00	2.20	4.90	5.00	9
10	MarkItDown	**3.64**	4.90	0.00	5.00	2.20	4.90	4.90	1

Note: Docling keeps the clause numbering best.

financial_statement_multiheader.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**4.86**	4.90	4.70	4.80	5.00	4.90	4.90	20
2	Docling	**4.75**	4.70	4.70	4.80	4.80	4.60	4.80	159
3	OpenDataLoader	**4.51**	4.90	4.80	5.00	2.50	4.90	4.90	15
4	pdfplumber	**4.41**	4.90	4.50	4.90	2.20	5.00	4.90	5

5	MarkItDown	**4.29**	5.00	3.30	5.00	2.50	5.00	5.00	1
6	PyMuPDF	**3.94**	4.30	3.60	4.80	2.10	4.90	3.90	4
7	hybrid_pdf	**3.94**	4.30	3.60	4.80	2.10	4.90	3.90	4
8	Unstructured	**3.54**	4.90	0.00	5.00	1.50	4.90	4.90	1
9	Tesseract OCR	**3.20**	4.40	0.00	3.80	2.20	4.10	4.80	5
10	smart_pdf	**2.96**	3.10	2.70	3.20	1.50	5.00	2.30	0

Note: Two-row merged header. PyMuPDF4LLM and Docling keep it together. Smart Extract with paid parsers fell to 2.96 because it copied the same numbers twice (see section 7).

form_grid_ac4.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF	**4.55**	4.90	2.50	5.00	5.00	4.90	4.90	1
2	hybrid_pdf (tied)	**4.55**	4.90	2.50	5.00	5.00	4.90	4.90	0
3	smart_pdf (tied)	**4.55**	4.90	2.50	5.00	5.00	4.90	4.90	0
4	OpenDataLoader (tied)	**4.52**	4.90	2.50	5.00	5.00	4.90	4.90	13
5	Docling	**4.43**	5.00	1.60	5.00	5.00	5.00	5.00	16
6	pdfplumber	**4.17**	5.00	2.50	5.00	2.50	5.00	5.00	1
7	PyMuPDF4LLM	**4.10**	4.60	1.20	4.70	5.00	4.70	4.50	5
8	Unstructured	**3.69**	4.90	0.00	5.00	2.50	4.90	4.90	0
9	Tesseract OCR	**3.62**	4.90	0.00	5.00	2.10	4.80	4.90	2
10	MarkItDown	**3.60**	4.90	0.00	5.00	2.00	4.90	4.90	1

Note: Easy for the PyMuPDF family. Nothing special.

invoice_complex.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**4.85**	5.00	4.80	5.00	4.50	5.00	5.00	22
2	Docling	**4.71**	5.00	4.40	5.00	3.90	4.90	5.00	28
3	PyMuPDF4LLM	**4.67**	4.80	3.90	4.90	4.80	5.00	4.70	20
4	PyMuPDF	**4.61**	4.90	4.00	5.00	4.10	5.00	4.80	3
5	hybrid_pdf	**4.61**	4.90	4.00	5.00	4.10	5.00	4.80	1
6	smart_pdf	**4.47**	4.60	4.00	5.00	4.00	5.00	4.30	0

7	pdfplumber	**4.38**	4.80	4.70	5.00	2.10	5.00	4.70	2
8	MarkItDown	**4.05**	4.90	2.10	5.00	2.40	5.00	4.90	1
9	Unstructured	**3.68**	4.90	0.00	5.00	2.30	4.90	4.90	1
10	Tesseract OCR	**2.62**	3.40	0.00	3.40	1.60	2.60	4.70	2

Note: OpenDataLoader is best. smart_pdf adds a description of the QR code that is not in the answer.

ledger_repeated_header.pdf (Tough documents, 3 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	105
2	OpenDataLoader (tied)	**4.95**	5.00	4.70	5.00	5.00	5.00	5.00	39
3	PyMuPDF	**4.86**	5.00	4.20	5.00	5.00	5.00	5.00	40
4	PyMuPDF4LLM	**4.86**	5.00	4.20	5.00	5.00	5.00	5.00	35
5	hybrid_pdf	**4.86**	5.00	4.20	5.00	5.00	5.00	5.00	23
6	smart_pdf	**4.86**	5.00	4.20	5.00	5.00	5.00	5.00	12
7	pdfplumber	**4.45**	5.00	4.20	5.00	2.50	5.00	5.00	47
8	MarkItDown	**3.74**	5.00	0.00	5.00	2.40	5.00	5.00	36
9	Unstructured	**3.66**	5.00	0.00	5.00	2.00	5.00	5.00	6
10	Tesseract OCR	**3.01**	3.90	0.10	4.00	1.90	3.20	4.90	20

Note: The header repeating on every page did not hurt anybody.

long_annual_report_20p.pdf (Tough documents, 19 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	114
2	Docling (tied)	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	116
3	OpenDataLoader	**4.89**	5.00	4.40	5.00	5.00	5.00	5.00	268
4	pdfplumber	**4.58**	5.00	5.00	5.00	2.50	5.00	5.00	136
5	smart_pdf	**4.48**	4.70	4.00	4.00	4.80	5.00	4.40	35
6	PyMuPDF	**4.46**	4.70	3.90	3.90	4.80	5.00	4.40	152
7	hybrid_pdf	**4.46**	4.70	3.90	3.90	4.80	5.00	4.40	79
8	Unstructured	**3.86**	5.00	0.00	5.00	3.20	5.00	5.00	16

9	MarkItDown	**3.73**	5.00	0.00	5.00	2.40	5.00	5.00	152
10	Tesseract OCR	**3.55**	4.90	0.00	4.20	2.50	4.80	4.90	141

Note: PyMuPDF4LLM and Docling are perfect (5.0).

mixed_table_chart_photo.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	smart_pdf	**3.90**	3.60	3.60	5.00	4.00	3.20	5.00	2.90	9
2	PyMuPDF4LLM	**3.55**	4.50	3.60	0.00	4.20	3.70	4.60	4.30	4
3	OpenDataLoader	**3.51**	4.50	3.90	0.00	3.70	3.50	4.10	5.00	77
4	Docling	**3.44**	4.40	3.30	0.00	3.70	3.80	4.10	4.90	14
5	PyMuPDF	**3.35**	4.20	3.10	0.00	3.90	3.80	4.10	4.30	0
6	hybrid_pdf	**3.35**	4.20	3.10	0.00	3.90	3.80	4.10	4.30	0
7	MarkItDown	**3.33**	4.50	3.90	0.00	3.70	2.20	4.10	5.00	1
8	pdfplumber	**3.08**	4.30	3.30	0.00	3.70	1.70	4.10	4.50	1
9	Unstructured	**2.79**	4.40	0.00	0.00	3.70	2.60	4.00	4.90	0
10	Tesseract OCR	**2.09**	3.40	0.00	0.00	2.10	1.70	2.70	4.70	4

Note: Table + chart + photo on one page. smart_pdf is best because it reads the chart.

policy_schedule_nested.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**4.77**	5.00	3.60	5.00	5.00	5.00	5.00	58
2	PyMuPDF4LLM (tied)	**4.77**	5.00	3.60	5.00	5.00	5.00	5.00	2
3	MarkItDown	**4.58**	5.00	5.00	5.00	2.50	5.00	5.00	1
4	pdfplumber	**4.36**	5.00	3.60	5.00	2.50	5.00	5.00	0
5	Tesseract OCR	**3.67**	5.00	0.00	5.00	2.20	4.90	5.00	2
6	Docling	**3.62**	3.60	2.70	3.20	4.30	5.00	2.80	13
7	Unstructured	**3.58**	5.00	0.00	5.00	1.50	5.00	5.00	0
8	PyMuPDF	**2.92**	2.70	1.90	2.20	3.80	5.00	1.90	1
9	hybrid_pdf	**2.92**	2.70	1.90	2.20	3.80	5.00	1.90	0
10	smart_pdf	**2.92**	2.70	1.90	2.20	3.80	5.00	1.90	0

Note: A table inside a table. PyMuPDF4LLM and OpenDataLoader keep it right. PyMuPDF-based parsers repeat the outer text across columns.

quality_manual.docx (Tough documents)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Unstructured	**4.86**	4.90	4.80	4.60	5.00	5.00	4.90	0
2	MarkItDown	**4.40**	4.50	4.80	3.90	4.30	4.30	4.60	4
3	Docling	**4.19**	4.20	4.30	3.60	4.60	4.20	4.20	1
4	native Word reader	**4.09**	4.20	4.30	3.20	4.60	4.20	4.20	3

Note: Unstructured is clearly better than the native Word reader (the native one drops the page header).

scanned_financial_statement.pdf (Tough documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**4.49**	4.80	4.00	4.80	3.70	4.80	4.80	11
2	Docling (tied)	**4.41**	4.40	4.30	4.80	4.20	4.70	4.10	29
3	Unstructured	**3.03**	4.20	0.00	3.70	1.90	3.90	4.50	20
4	hybrid_pdf	**2.36**	3.30	0.00	2.40	1.30	2.50	4.80	12
5	Tesseract OCR	**2.36**	3.30	0.00	2.40	1.30	2.50	4.80	7
6	smart_pdf	**2.36**	3.30	0.00	2.40	1.30	2.50	4.80	8
-	pdfplumber	0.00 (empty)	-	-	-	-	-	-	0
-	PyMuPDF	0.00 (empty)	-	-	-	-	-	-	0
-	MarkItDown	0.00 (empty)	-	-	-	-	-	-	1
-	OpenDataLoader	0.00 (empty)	-	-	-	-	-	-	23

Note: A scan: parsers without OCR return nothing. Docling and PyMuPDF4LLM read it.

annual_report_income_statement.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**4.77**	4.90	4.30	5.00	4.50	4.90	4.90	4
2	PyMuPDF4LLM (tied)	**4.71**	4.90	3.50	5.00	5.00	4.90	4.90	5
3	Docling	**4.60**	4.70	4.10	4.60	4.80	5.00	4.40	32
4	pdfplumber	**4.36**	4.90	4.20	4.90	2.10	5.00	4.90	0
5	MarkItDown	**4.34**	5.00	3.50	5.00	2.50	5.00	5.00	0

6	PyMuPDF	**4.33**	4.40	3.70	4.50	4.60	4.90	3.90	0
7	hybrid_pdf	**4.33**	4.40	3.70	4.50	4.60	4.90	3.90	0
8	smart_pdf	**4.33**	4.40	3.70	4.50	4.60	4.90	3.90	1
9	Unstructured	**3.62**	4.90	0.00	5.00	1.90	4.90	4.90	10
10	Tesseract OCR	**2.32**	3.20	0.00	1.80	1.90	2.40	4.60	1

Note: Dollar signs in their own cells and numbers in brackets. Layout-aware parsers (OpenDataLoader, PyMuPDF4LLM, Docling) beat plain PyMuPDF by 0.3 to 0.4.

bank_statement.pdf (Real-world style documents, 3 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**4.77**	5.00	5.00	5.00	3.60	5.00	5.00	4
2	OpenDataLoader (tied)	**4.76**	4.90	4.90	4.90	4.00	4.80	5.00	5
3	Docling (tied)	**4.71**	4.90	5.00	4.90	3.70	4.80	5.00	35
4	PyMuPDF (tied)	**4.68**	4.80	4.70	4.90	3.90	4.90	4.80	2
5	hybrid_pdf (tied)	**4.68**	4.80	4.70	4.90	3.90	4.90	4.80	1
6	smart_pdf (tied)	**4.68**	4.80	4.70	4.90	3.90	4.90	4.80	2
7	pdfplumber	**4.54**	5.00	5.00	5.00	2.20	5.00	5.00	1
8	MarkItDown	**3.98**	5.00	1.60	5.00	2.30	5.00	5.00	2
9	Unstructured	**3.57**	5.00	0.00	5.00	1.40	5.00	5.00	2
10	Tesseract OCR	**2.99**	4.00	0.00	3.80	1.90	3.40	4.80	4

Note: 3 pages, header repeated. Every parser with a text layer does well (4.5 to 4.8).

bill_of_lading.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**4.56**	4.80	3.60	4.60	4.90	4.50	5.00	5
2	Docling	**4.45**	4.50	3.40	4.80	4.80	5.00	4.20	7
3	PyMuPDF4LLM	**4.38**	4.50	3.30	4.60	4.80	4.40	4.70	2
4	PyMuPDF	**3.85**	4.00	2.30	4.00	4.50	4.60	3.60	0
5	hybrid_pdf	**3.85**	4.00	2.30	4.00	4.50	4.60	3.60	0
6	smart_pdf	**3.85**	4.00	2.30	4.00	4.50	4.60	3.60	0
7	pdfplumber	**3.84**	4.60	2.80	4.60	2.00	4.50	4.60	0

8	MarkItDown	**3.66**	4.80	1.10	4.60	2.00	4.50	5.00	0
9	Unstructured	**3.46**	4.80	0.00	4.60	1.90	4.50	5.00	0
10	Tesseract OCR	**3.30**	4.50	0.00	4.20	1.90	4.20	4.90	1

Note: Merged cells. OpenDataLoader, Docling and PyMuPDF4LLM are ahead of the PyMuPDF family (3.85).

board_minutes.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	2
2	Docling (tied)	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	7
3	PyMuPDF (tied)	**4.91**	4.90	4.80	5.00	5.00	4.90	4.90	0
4	hybrid_pdf (tied)	**4.91**	4.90	4.80	5.00	5.00	4.90	4.90	0
5	smart_pdf (tied)	**4.91**	4.90	4.80	5.00	5.00	4.90	4.90	0
6	OpenDataLoader	**4.61**	4.90	4.80	5.00	3.20	4.90	4.90	3
7	pdfplumber	**4.56**	5.00	5.00	5.00	2.30	5.00	5.00	0
8	Unstructured	**3.69**	4.90	0.00	5.00	2.40	4.90	4.90	0
9	MarkItDown	**3.65**	4.90	0.00	5.00	2.20	4.90	4.90	0
10	Tesseract OCR	**3.64**	4.80	0.00	4.90	2.40	4.70	5.00	1

Note: Mostly text. PyMuPDF4LLM and Docling are perfect. By document picked OpenDataLoader (4.61) because its rule is per document type, not per file.

hr_headcount_report.xlsx (Real-world style documents)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	Unstructured	**3.75**	3.60	3.50	5.00	3.70	3.10	3.20	4.00	1
2	native Excel reader (tied)	**3.73**	3.70	3.50	5.00	3.70	2.90	3.30	4.10	0
3	Docling	**3.56**	3.50	3.80	5.00	3.80	1.50	4.20	3.00	0
4	MarkItDown	**3.40**	3.30	2.80	5.00	3.40	2.70	3.50	3.10	0

Note: Excel with a real chart object. Native reader and Unstructured are tied (3.7).

lab_report.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**4.93**	5.00	4.60	5.00	5.00	5.00	5.00	10

2	PyMuPDF4LLM (tied)	**4.88**	4.90	4.40	5.00	5.00	5.00	4.90	2
3	PyMuPDF (tied)	**4.85**	4.90	4.50	5.00	4.90	5.00	4.80	0
4	hybrid_pdf (tied)	**4.85**	4.90	4.50	5.00	4.90	5.00	4.80	0
5	smart_pdf (tied)	**4.85**	4.90	4.50	5.00	4.90	5.00	4.80	0
6	OpenDataLoader	**4.55**	4.70	4.30	5.00	4.00	5.00	4.40	4
7	pdfplumber	**4.38**	5.00	4.50	5.00	1.90	5.00	4.90	0
8	MarkItDown	**4.26**	5.00	3.00	5.00	2.50	5.00	5.00	0
9	Unstructured	**3.68**	4.90	0.00	5.00	2.20	5.00	4.90	0
10	Tesseract OCR	**3.02**	4.00	0.00	3.90	2.20	3.80	4.30	1

Note: Docling, PyMuPDF4LLM and the PyMuPDF family all above 4.8. By part scored only 3.26: the table piece came out with every merged cell repeated across 12 columns.

payslip.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**4.69**	5.00	4.00	5.00	4.20	5.00	5.00	10
2	PyMuPDF4LLM (tied)	**4.62**	5.00	4.00	5.00	3.70	5.00	5.00	2
3	MarkItDown	**4.34**	5.00	4.00	5.00	2.00	5.00	5.00	0
4	PyMuPDF	**4.05**	4.90	0.90	5.00	3.70	5.00	4.80	0
5	hybrid_pdf	**4.05**	4.90	0.90	5.00	3.70	5.00	4.80	0
6	smart_pdf	**4.05**	4.90	0.90	5.00	3.70	5.00	4.80	0
7	OpenDataLoader	**3.94**	5.00	0.00	5.00	3.70	5.00	5.00	2
8	pdfplumber	**3.67**	4.90	0.90	5.00	1.40	5.00	4.80	0
9	Unstructured	**3.61**	5.00	0.00	5.00	1.70	5.00	5.00	0
10	Tesseract OCR	**3.53**	4.80	0.00	4.80	1.90	4.60	5.00	1

Note: Two tables side by side. Docling and PyMuPDF4LLM are best (4.6 to 4.7); OpenDataLoader, the choice for this type, scored 3.94.

purchase_order.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**4.96**	5.00	4.80	5.00	5.00	5.00	5.00	4
2	PyMuPDF	**4.79**	4.90	4.50	5.00	4.40	5.00	4.90	0
3	PyMuPDF4LLM	**4.79**	4.70	4.60	5.00	4.90	4.90	4.60	8

4	hybrid_pdf	**4.79**	4.90	4.50	5.00	4.40	5.00	4.90	0
5	Docling	**4.79**	5.00	5.00	5.00	3.80	5.00	5.00	11
6	smart_pdf	**4.79**	4.90	4.50	5.00	4.40	5.00	4.90	0
7	pdfplumber	**4.30**	4.90	4.50	5.00	1.50	5.00	4.90	0
8	MarkItDown	**4.02**	5.00	1.90	5.00	2.20	5.00	5.00	0
9	Unstructured	**3.80**	5.00	0.00	5.00	2.80	5.00	5.00	0
10	Tesseract OCR	**3.06**	4.20	0.00	3.80	1.90	3.70	4.70	1

Note: OpenDataLoader 4.96 is best. By part scored only 3.54.

quarterly_review.pptx (Real-world style documents)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**4.50**	4.40	4.70	5.00	5.00	3.70	3.90	4.90	0
2	MarkItDown	**4.12**	4.70	3.60	5.00	2.90	3.30	4.70	4.60	0
3	native PowerPoint reader	**4.01**	4.20	3.50	3.80	3.90	4.20	3.70	4.90	0
4	Unstructured	**3.72**	3.70	3.50	3.80	3.90	3.30	2.90	5.00	0

Note: Deck with a real chart. Docling reads the chart (4.50); the native reader does not (4.01).

research_two_column.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	smart_pdf	**4.43**	4.50	3.70	5.00	4.80	4.00	5.00	4.10	4
2	Docling	**3.79**	4.80	3.80	0.00	4.00	4.40	4.60	4.90	8
3	PyMuPDF4LLM	**3.74**	4.70	3.10	0.00	4.30	4.60	4.80	4.70	7
4	OpenDataLoader	**3.72**	4.70	3.80	0.00	4.00	4.10	4.50	5.00	4
5	PyMuPDF	**3.50**	4.50	2.60	0.00	4.20	4.10	4.50	4.60	0
6	hybrid_pdf	**3.50**	4.50	2.60	0.00	4.20	4.10	4.50	4.60	0
7	pdfplumber	**3.24**	4.60	2.90	0.00	4.00	1.90	4.50	4.80	0
8	Tesseract OCR	**2.57**	4.40	0.00	0.00	2.70	2.20	4.30	4.40	1
9	MarkItDown	**2.23**	3.60	0.00	0.00	2.20	2.40	3.10	4.20	1
10	Unstructured	**2.14**	3.40	0.00	0.00	2.20	2.10	2.80	4.40	0

Note: Two columns with a chart. smart_pdf is clearly best (4.43) because it reads the chart.

rfp_requirements_matrix.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	10
2	PyMuPDF (tied)	**4.95**	5.00	4.90	5.00	5.00	5.00	4.90	0
3	hybrid_pdf (tied)	**4.95**	5.00	4.90	5.00	5.00	5.00	4.90	0
4	smart_pdf (tied)	**4.95**	5.00	4.90	5.00	5.00	5.00	4.90	0
5	PyMuPDF4LLM	**4.89**	4.90	4.70	5.00	4.90	4.80	5.00	3
6	OpenDataLoader	**4.75**	4.90	4.70	5.00	4.10	4.80	5.00	3
7	pdfplumber	**4.40**	4.90	4.60	5.00	2.20	4.80	4.90	0
8	MarkItDown	**3.66**	4.90	0.00	5.00	2.30	4.80	5.00	1
9	Unstructured	**3.66**	4.90	0.00	5.00	2.20	4.80	5.00	0
10	Tesseract OCR	**3.64**	4.90	0.00	5.00	2.20	4.70	5.00	1

Note: Rowspan category cells and check boxes were easy: every top parser is 4.75 or better.

scanned_purchase_order_phone.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	Docling	**4.67**	5.00	3.30	5.00	4.80	5.00	4.90	13
2	PyMuPDF4LLM	**3.58**	4.50	0.50	4.00	3.60	4.50	4.40	7
3	Unstructured	**2.90**	3.80	0.00	3.30	2.50	3.30	4.50	9
4	hybrid_pdf	**2.30**	3.20	0.00	2.30	1.60	2.80	3.80	2
5	Tesseract OCR	**2.30**	3.20	0.00	2.30	1.60	2.80	3.80	2
6	smart_pdf	**2.30**	3.20	0.00	2.30	1.60	2.80	3.80	5
-	pdfplumber	0.00 (empty)	-	-	-	-	-	-	0
-	PyMuPDF	0.00 (empty)	-	-	-	-	-	-	0
-	MarkItDown	0.00 (empty)	-	-	-	-	-	-	0
-	OpenDataLoader	0.00 (empty)	-	-	-	-	-	-	2

Note: A tilted, noisy phone photo. Docling 4.67 is far ahead; hybrid_pdf and smart_pdf (Tesseract only) get 2.30.

sop_goods_receipt.docx (Real-world style documents)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds

1	Unstructured	**4.93**	4.90	4.80	5.00	5.00	4.90	5.00	0
2	Docling	**4.57**	4.60	5.00	3.90	4.80	4.20	5.00	0
3	MarkItDown	**4.34**	4.40	4.80	3.70	4.30	4.10	4.80	1
4	native Word reader	**4.33**	4.50	5.00	2.80	4.80	4.10	4.90	0

Note: Unstructured (4.93) is ahead of Docling (4.57) and the native Word reader (4.33).

utility_bill.pdf (Real-world style documents, 1 pages)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	smart_pdf	**4.49**	4.60	4.50	5.00	4.90	3.10	5.00	4.30	4
2	PyMuPDF4LLM	**3.58**	4.70	3.20	0.00	4.30	3.40	4.80	4.60	2
3	OpenDataLoader	**3.49**	4.50	3.00	0.00	4.30	3.50	4.10	5.00	3
4	Docling	**3.45**	4.60	3.20	0.00	4.30	2.80	4.20	5.00	9
5	PyMuPDF	**3.44**	4.40	2.90	0.00	4.30	3.50	4.10	4.90	0
6	hybrid_pdf	**3.44**	4.40	2.90	0.00	4.30	3.50	4.10	4.90	0
7	pdfplumber	**3.19**	4.40	2.90	0.00	4.30	1.70	4.10	4.90	0
8	MarkItDown	**2.84**	4.50	0.00	0.00	4.30	2.00	4.10	5.00	1
9	Unstructured	**2.73**	4.50	0.00	0.00	4.30	1.20	4.10	5.00	0
10	Tesseract OCR	**2.70**	4.50	0.00	0.00	3.80	1.70	4.40	4.60	1

Note: Bill with a bar chart. smart_pdf wins (4.49) because it reads the chart; the free route scores 3.58.

annual_report_60p.pdf (Long documents, 56 pages)

Rank	Parser	Score	Text	Table	Chart	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**4.18**	5.00	4.80	0.00	4.90	4.70	5.00	5.00	52
2	OpenDataLoader	**3.91**	4.90	4.10	0.00	4.30	4.30	4.80	5.00	20
3	pdfplumber	**3.86**	5.00	4.80	0.00	4.80	2.40	5.00	5.00	18
4	MarkItDown	**3.56**	5.00	2.80	0.00	4.80	2.40	5.00	5.00	13
5	PyMuPDF	**3.46**	4.40	3.00	0.00	3.30	4.40	5.00	4.00	24
6	hybrid_pdf	**3.46**	4.40	3.00	0.00	3.30	4.40	5.00	4.00	25
7	Unstructured	**3.17**	5.00	0.00	0.00	4.80	2.40	5.00	5.00	22
8	Tesseract OCR	**3.12**	4.90	0.00	0.00	4.70	2.40	4.90	5.00	144

-	Docling	0.00 (timeout)	-	-	-	-	-	-	-	600

big_workbook.xlsx (Long documents)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	native Excel reader	**4.99**	5.00	5.00	5.00	5.00	5.00	5.00	13
2	MarkItDown	**4.82**	4.80	4.70	4.70	5.00	4.80	4.80	19
3	Docling	**4.58**	5.00	5.00	5.00	2.50	5.00	5.00	41
4	Unstructured	**4.56**	5.00	4.90	5.00	2.50	5.00	5.00	18

contract_45p.pdf (Long documents, 46 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	PyMuPDF4LLM	**4.99**	5.00	-	5.00	4.90	5.00	5.00	22
2	PyMuPDF (tied)	**4.90**	5.00	-	5.00	4.50	5.00	5.00	14
3	hybrid_pdf (tied)	**4.90**	5.00	-	5.00	4.50	5.00	5.00	14
4	Docling	**4.88**	4.90	-	4.60	5.00	4.90	5.00	176
5	OpenDataLoader	**4.79**	4.90	-	4.60	4.60	4.90	5.00	11
6	pdfplumber	**4.49**	5.00	-	5.00	2.40	5.00	5.00	16
7	MarkItDown	**4.49**	5.00	-	5.00	2.40	5.00	5.00	14
8	Tesseract OCR	**4.49**	5.00	-	5.00	2.40	5.00	5.00	196
9	Unstructured	**4.49**	5.00	-	5.00	2.40	5.00	5.00	10

handbook_40p.docx (Long documents)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	native Word reader	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	5
2	Unstructured (tied)	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	6
3	Docling (tied)	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	14
4	MarkItDown (tied)	**4.99**	5.00	5.00	5.00	5.00	5.00	5.00	6

scanned_report_30p.pdf (Long documents, 31 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds

1	Docling	**4.58**	4.50	4.70	4.80	4.50	4.50	4.50	387
2	PyMuPDF4LLM	**3.67**	3.90	2.50	4.10	3.60	3.70	4.20	138
3	Unstructured	**3.02**	4.10	0.00	1.10	4.40	3.50	5.00	160
4	hybrid_pdf	**2.66**	4.20	0.00	1.10	2.10	3.60	5.00	74
5	Tesseract OCR	**2.66**	4.20	0.00	1.10	2.10	3.60	5.00	124
-	pdfplumber	0.00 (empty)	-	-	-	-	-	-	0
-	PyMuPDF	0.00 (empty)	-	-	-	-	-	-	0
-	MarkItDown	0.00 (empty)	-	-	-	-	-	-	0
-	OpenDataLoader	0.00 (empty)	-	-	-	-	-	-	1

transaction_ledger_100p.pdf (Long documents, 100 pages)

Rank	Parser	Score	Text	Table	Numbers	Structure	Complete	No extras	Seconds
1	OpenDataLoader	**5.00**	5.00	5.00	5.00	5.00	5.00	5.00	13
2	PyMuPDF4LLM (tied)	**4.99**	5.00	4.90	5.00	5.00	5.00	5.00	62
3	pdfplumber	**4.58**	5.00	5.00	5.00	2.50	5.00	5.00	40
4	MarkItDown	**3.60**	5.00	0.00	5.00	1.60	5.00	5.00	47
5	Unstructured	**3.60**	5.00	0.00	5.00	1.60	5.00	5.00	55
6	PyMuPDF	**3.01**	2.60	2.70	2.60	3.40	5.00	1.70	294
7	hybrid_pdf	**3.01**	2.60	2.70	2.60	3.40	5.00	1.70	292
8	Tesseract OCR	**2.10**	2.50	0.00	2.60	1.30	1.80	4.40	160